Government Digital Procurement, Cost Structure, and Firm Innovative Economic Efficiency

Yupeng Zhang[a], Bingxin Zheng[a,*]

[a] *East China Normal University, School of Economics and Management,200062, Shanghai, China*

Abstract

This study investigates how government procurement of digital products and services affects firms' efficiency in converting R&D expenditure into economic revenues. Leveraging the BERT-wwm text classification algorithm, we identify digital procurement contracts from a comprehensive Chinese procurement dataset. We then employ a random coefficient model with rolling-window estimation to construct an innovative economic efficiency indicator, measured as the revenue elasticity of R&D expenditure. The results show that while digital procurement stimulates R&D investment, it significantly erodes firms' innovative economic efficiency. Mechanism analysis indicates that digital procurement induces more flexible cost structures, thereby undermining the irreversible investments vital for sustainable innovation. The negative effect is stronger for state-owned enterprises and repeat contract winners but weaker among firms with high initial asset rigidity, operating in more competitive markets, and in regions with superior institutional quality. These findings advance the literature on government procurement and firm innovation and offer policy insights for optimizing procurement design in China.

Keywords: Government digital procurement, innovative economic

*Corresponding author.

Email addresses: ypzhang@sfs.ecnu.edu.cn (Yupeng Zhang), bxzheng0510@outlook.com (Bingxin Zheng)

efficiency, cost structure, China

1. Introduction

The role of public procurement is evolving beyond mere cost-containment toward serving as a strategic instrument for societal value creation (Grandia and Volker, 2023). This shift enables public entities to leverage their procurement scale to redirect innovative capital to frontier technologies (Edler and Georghiou, 2007; OECD, 2017; Uyarra et al., 2020). Amidst the digital economy's rise as a global growth engine,[1] China has institutionalized digital transformation within its national strategic framework, as exemplified by the *14th Five-Year Plan for Digital Economy Development*. Accordingly, China's public procurement has progressively expanded its scope to strategic digital sectors, using public demand to internalize learning effects and economies of scale in the adoption of frontier technologies.[2]

While this transition may fundamentally alter corporate R&D trajectories, quantifying these effects requires a reliable measure of digital procurement intensity. The primary empirical obstacle is that digital procurement is not driven by explicit policy events, which precludes the use of quasi-experimental research designs. Furthermore, the lack of a standardized definition for this specialized category makes disentangling digital from conventional procurement empirically challenging. To address these hurdles, we leverage China's official digital economy classification standards[3] with a

[1]International organizations estimate that digital activities already account for around 15% of global GDP, and that this share has been growing significantly faster than the real economy as digital technologies diffuse across countries and sectors (United Nations Conference on Trade and Development, 2019; World Bank, 2023, 2025).

[2]Data released by the *Ministry of Finance* indicate that China's national government procurement in 2022 amounted to approximately 3.5 trillion RMB, equivalent to 2.9% of GDP and 9.4% of national fiscal expenditure. Based on our sample, the value of government digital procurement contracts awarded to listed companies reached RMB 17.8 billion in 2022 alone, after adjusting for outliers.

[3]*The Statistical Classification of the Digital Economy and Its Core Industries (2021)*, issued by the National Bureau of Statistics, provides the official framework for defining the scope of digital economy activities in China and therefore offers an authoritative basis

fine-tuned Chinese BERT-wwm classification algorithm. Specially, we first link more than 3.6 million contract-level records from the China Government Procurement Network to publicly listed firms. We then fine-tune the classification algorithm using 5,000 manually labeled samples (achieving 94% accuracy) to identify digital economy–related contracts. Finally, we aggregate these predictions to the firm-year level to construct the government digital procurement indicator[4].

We then examine the impact of government digital procurement on corporate innovation and its resulting economic performance.While established studies confirm that public procurement generally stimulates innovation (Dai et al., 2021; Gross and Sampat, 2023; Lichtenberg, 1988; Slavtchev and Wiederhold, 2016), the efficacy of such policy tools depends on contract design. Howell et al. (2021) highlight this critical distinction within the U.S. defense sector: open competitions–where the buyer specifies a problem and allows suppliers to propose solutions–encourage commercial innovation, while highly specified contracts tend to increase program lock-in without yielding technological gains. Similarly, Krieger et al. (2024) find that rigid, price-based contracts reduce product and process innovation while redirecting firms' efforts toward established offerings. Furthermore, Kantor and Whalley (2025) examine the economic impact of Space Race public R&D spending, finding that space-related industries experienced increased patenting, while only moderate gains in manufacturing value-added that were no larger than those of non-R&D government expenditures. This suggests a disconnect between innovation activities and the actual economic impact. Government digital procurement often mirrors the restrictive contract features of these studies, as it is typically customized, compliance-intensive, and mission-oriented. Given this structural rigidity, such contracts may reward

for identifying and measuring government digital procurement in this paper.

[4]In this paper, government digital procurement refers to the procurement of digital products and services, rather than the digitization of procurement procedures or digital government construction.

unscalable, project-specific solutions, thereby making it unclear whether digital procurement truly improves innovative economic efficiency or simply increases observable innovation inputs.

To address this concern, we employ the Research Quotient (RQ) indicator proposed by Cooper et al. (2022) as our primary measure of innovative economic efficiency. RQ is defined as the elasticity of firm revenue with respect to the stock of R&D capital, which captures the entire efficiency of transforming R&D inputs into innovative outputs (whether patented or not) and ultimately into economic value. We construct RQ for Chinese listed firms by estimating a firm-level production function that includes R&D capital, physical capital, labor, and related inputs. Employing a dataset of Chinese A-share listed firms from 2015 to 2023, we document a robust yet counter-intuitive pattern: while digital procurement stimulates innovation investment, it simultaneously undermines innovative economic efficiency. In our preferred specifications with comprehensive fixed effects, a one-standard-deviation increase in log-transformed digital procurement intensity is associated with a 1.02 percentage-point standard-deviation decline in RQ.

To further clarify the underlying mechanism, we unpack the black box of firm cost structure. Consistent with management accounting literature (Banker et al., 2024a,b), we treat cost structure–the mix of fixed versus variable costs–as firm's strategic choice, where a higher proportion of variable costs denotes flexibility and high fixed costs signal rigidity. Drawing on the cost behavior model of Banker et al. (2014)[5] and a mediation analysis framework following Zhang and Yang (2025), we uncover a two-stage mediating process. In the first stage, strict delivery and customization requirements in digital procurement force suppliers to transition from fixed-cost commitments to variable-cost operations. Our analysis confirms that government

[5]Banker et al. (2014) define cost flexibility (or rigidity) as the percentage change in costs in response to a one-percent change in sales. Firms with a higher ratio of variable to fixed costs exhibit greater cost flexibility, as their costs adjust more significantly to changes in sales.

digital procurement significantly enhances cost flexibility. In the second stage, this heightened flexibility erodes the hard commitments required to pursue a product-leadership strategy (Banker et al., 2024a), thereby impairing innovative value creation. This inference is evidenced by a robust negative relationship between procurement-induced cost flexibility and RQ. Together, our findings suggest that by adopting a more flexible cost structure to accommodate idiosyncratic public demand, firms inadvertently sacrifice their capacity for sustained innovation.

A major concern regarding our previous results is that government digital procurement may be endogenous. Since successfully securing government contracts is not an exogenous or randomized treatment, firms with weaker innovation prospects may actively seek government orders as a substitute for insufficient market demand. To mitigate this concern, we implement three instrumental variable (IV) strategies. First, we construct the annual industrial average of digital procurement intensity to capture annual common demand shocks within each technological segment. Second, following the logic of shift-share instruments (Bartik, 1991), we construct a Bartik instrument by interacting base-period industrial-level shares of digital procurement with the subsequent nationwide growth in digital procurement, calculated by excluding the contribution of the firm's own industry and region. Third, we employ the heteroskedasticity-based IV method proposed by Lewbel (2012), which generates internal instruments using the product of the centered exogenous covariates and the first-stage residuals. Across these strategies, the IV estimates consistently support the finding that government digital procurement exerts a significantly negative causal impact on firms' innovative economic efficiency by inducing a more flexible cost structure. The results remain robust under alternative model specifications. These include models that address simultaneity by lagging digital procurement, as well as the use of high-dimensional fixed effects, alternative constructions of RQ based on varying rolling windows or stock-versus flow-based measures, and alternative clustering of standard errors.

Our analysis also reveals pronounced heterogeneity: the negative link between digital procurement and innovation efficiency is amplified for state-owned enterprises (SOEs) and repeat contract winners. In contrast, the negative effect is significantly mitigated among firms with higher initial asset rigidity, in more competitive industries, and in provinces with better-developed legal and intermediary institutions.

Related literature. This study contributes to the growing literature on government procurement and firm innovation. While a substantial body of research demonstrates that government procurement–including innovation-oriented, green, and medical procurement–significantly shapes firm innovation outcomes (Clemens and Rogers, 2026; Krieger et al., 2024; Krieger and Zipperer, 2022), the specific implications of digital procurement remain under-explored. To date, the few existing studies have primarily documented its role in driving digital transformation (Chen et al., 2025) and increasing innovation volume (Tan et al., 2024), yet largely overlook the economic efficiency of such innovation.

Theoretically, public procurement–especially when it involves technologically sophisticated demand–can stimulate firm innovation by reducing demand uncertainty and generating certification and learning effects (Aschhoff and Sofka, 2009; Guerzoni and Raiteri, 2015; Slavtchev and Wiederhold, 2016). Yet this positive effect is not guaranteed. Procurement efficacy depends critically on contract design and institutional implementation, as rigid contractual constraints or political distortions can dampen incentives for valuable innovation (Choi et al., 2024; Krieger et al., 2024; Tang et al., 2025). This issue is particularly salient in the digital procurement context, where project-based procurement often prioritizes rapid technological deployment over long-term value creation. Furthermore, given that supply-side policies (e.g., subsidies) have been shown to distort incentives by generating patent bubbles without corresponding gains in market competitiveness (Cheng et al., 2019; Hu et al., 2017; Zhang and Yang, 2025), it remains unclear whether demand-side instruments induce similar inefficiencies. Our study sheds light on these issues, revealing that while digital procurement significantly boosts innovation investment, it simultaneously erodes firms' innovative economic efficiency. This finding indicates that digital procurement can trigger a divergence between innovation volume and its economic substance.

Second, we contribute to the cost accounting literature by identifying cost structure as a pivotal mechanism linking government intervention to innovation efficiency. Prior studies examine how government interventions–such as healthcare regulation, judicial efficiency and shareholder protection, and employment protection laws–shape firms' cost structures (Banker et al., 2013a,b; Belina et al., 2019; Holzhacker et al., 2015; Kallapur and Eldenburg, 2005), while another stream documents the consequences of cost structure for earnings predictability, dividend policy, income smoothing, and tax-motivated income shifting (Banker and Chen, 2006; Hartlieb and Loy, 2022; He et al., 2020; McGuire et al., 2023). However, few studies integrate these two perspectives within a unified framework that connects the determinants of cost structure to its downstream implications for innovation outcomes.

The study most closely related to ours is Zhang and Yang (2025), who find that innovation subsidies reduce innovative economic efficiency by increasing cost rigidity. In contrast, we show that government digital procurement also reduces innovative economic efficiency, but through increasing cost flexibility. We conjecture that these seemingly divergent findings can be reconciled by the existence of an optimal cost structure that maximizes innovation outcomes. Government interventions that push firms away from this optimum–toward either excessive rigidity or excessive flexibility–undermine innovative efficiency. Innovation subsidies encourage greater fixed-asset investment, thereby increasing cost rigidity, whereas the inherent characteristics of government digital procurement increase cost flexibility. Both forms of deviation impair innovation performance. Overall, our results highlight the critical importance of preserving an optimal cost structure to ensure that government interventions effectively foster economically valuable innovation.

Third, we uncover the organizational and institutional boundaries of the procurement-innovation relationship. Our analysis reveals that the efficiency of demand-driven innovation depends heavily on firm-level characteristics and market-wide frictions. Specifically, this efficiency loss is amplified for SOEs and repeat contract winners, consistent with the idea that multi-objective mandates and relational lock-ins intensify the distortionary effects of public demand (Arora et al., 2025; Macchiavello and Morjaria, 2023; Shleifer and Vishny, 1994; Wang et al., 2024). In contrast, the negative effect is attenuated for firms with high initial asset rigidity, as well as those operating under intense product-market competition or superior institutional environments. These internal and external structural constraints curb excessive contract-specific customization and instead induce firms to codify procurement-driven projects into scalable and redeployable innovations.

The subsequent sections are structured as follows. Section 2 describes the institutional setting of government procurement in China. Section 3 lays out the theoretical foundation and hypothesis development. Section 4 describes the methodology, detailing the data construction for government digital procurement as well as the estimation of RQ. Section 5 analyzes the baseline effects on innovative economic efficiency and explores the mediating role of cost structures. Section 6 performs a series of robustness checks. Section 7 presents multi-dimensional heterogeneity analyses. Section 8 concludes and discusses policy implications.

2. Government procurement in China

China's government procurement regime has gradually shifted from administrative allocation toward a more rule-based system. *The Government Procurement Law of the People's Republic of China* (effective in 2003) and its supporting *Regulation on the Implementation of the Government Procurement Law* (effective in 2015) provide a unified legal framework for purchasing goods and services with fiscal funds.[6] This framework is grounded in principles such as transparency, fair competition, procedural integrity, and good faith. Around 2015, the aforementioned implementing regulation and subsequent disclosure rules significantly reshaped the procurement environment. Procurement announcements, award information, and contract-related disclosures became more systematically institutionalized and consolidated through official publication channels, which makes contract-level empirical analysis feasible at scale.

China's procurement landscape is characterized by a fundamental tension between centralized regulatory frameworks and decentralized implementation. While central rule-making ensures formal uniformity, decentralized implementation still leads to significant divergence in enforcement intensity, procurement expertise, and discretionary practices across local governments.[7] Local preferences are often embedded in tender specifications or evaluation criteria—for instance, through clauses prioritizing local suppliers or de facto entry barriers, such as mandatory local registration, which disadvantage external competitors (Tang et al., 2025). Such frictions ultimately undermine

---

[6] *Government Procurement Law of the People's Republic of China*(`https://www.ndrc.gov.cn/xxgk/zcfb/qt/200507/t20050706_967929.html`); *Regulation on the Implementation of the Government Procurement Law* (`https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzxzfg/201502/t20150227_350612.html`).

[7] Article 5 of *Regulations for the Implementation of the Government Procurement Law of the People's Republic of China* authorizes provinces, autonomous regions, and municipalities (or their authorized bodies) to set centralized procurement catalogues and procurement threshold standards for provincial, prefectural, and county levels (`https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzxzfg/201502/t20150227_350612.html`). In practice, local governments also issue platform- and procedure-specific implementation rules. See the Regulations for *Notice of the Beijing Municipal Finance Bureau on the Beijing Government Centralized Procurement Catalogue and Standards (2023 Edition)* (`https://www.beijing.gov.cn/zhengce/zhengcefagui/202301/t20230103_2890663.html`), and *Measures for the Management of the Shanghai Government Procurement Cloud Platform Electronic Marketplace* (`https://czj.sh.gov.cn/zys_8908/zcfg_8983/zcfb_8985/zfcg_9033/20250610/xxfboswf0000003286.html`).

the system's capacity to identify the most technically proficient solutions.

Against the broader state push to develop the digital economy, Chinese government procurement has increasingly incorporated digital products and services, including software, information systems, cloud services, data processing, and related digital infrastructures.[8] The Guideline on Building a Digital Government issued by the State Council (Guo Fa [2022] No. 14) explicitly mandates leveraging government procurement to foster digital technology innovation and standardize project construction.[9] This policy orientation is further reflected in the *Interim Measures for the Government Procurement Management of Government Information Systems*, which provide that, in principle, government information systems that do not involve state secrets and serve external users should adopt a cloud-computing model.[10]

Beyond the inherent challenges of conventional public procurement, digital procurement introduces distinctive features and complexities. Stringent mandates concerning data security, system reliability, and backward compatibility—coupled with an increasing dependence on standardized specifications—embed procurement processes within a rigid institutional framework.[11] When requirements are anchored to slow-evolving standards or nar-

[8] See the *14th Five-Year Digital Economy Development Plan* (https://www.ndrc.gov.cn/fggz/fzzlgh/gjjzxgh/202203/t20220325_1320207.html), which frames digital infrastructure, digital services, and digital government as central components of China' s digital-economy strategy.

[9] https://www.gov.cn/gongbao/content/2022/content_5699869.html.

[10] https://m.mof.gov.cn/czxw/201801/t20180102_2791414.htm.

[11] For example, the *Regulations for the Implementation of the Government Procurement Law of the People's Republic of China*, which require procurement needs to satisfy technical, service, and security requirements; and which also classify projects with unified technical or service standards as suitable for centralized procurement.(https://www.moj.gov.cn/pub/sfbgw/flfggz/flfggzxzfg/201502/t20150227_350612.html). *The Ministry of Finance' s Database Government Procurement Demand Standard (2023 Edition)* requires many public-sector database purchases to include compliance with security-and-reliability assessment requirements as a substantive procurement requirement.(https://gks.mof.gov.cn/guizhangzhidu/202312/t20231226_3924124.htm) The *Operating Sys-*

rowly defined catalogs, procurement frequently devolves into a conservative exercise in checklist-based compliance. This structural rigidity, aggravated by high ex post adaptation costs, precipitates significant bargaining frictions, and persistent vendor lock-in. Consequently, these frictions predispose firms toward bespoke customization, which, while intensifying demand-driven innovation, ultimately constrains the redeployability and broader technological value of these solutions.

In sum, China’s procurement institutions combines centralized rule-making, decentralized implementation, and a growing reliance on standards-based digital purchasing within a rigid contractual framework. This institutional backdrop motivates our analysis in Section 3, clarifying why greater exposure to digital procurement may not necessarily generate substantial innovation revenues.

3. Hypotheses development

3.1. Government digital procurement and innovative economic efficiency

Government digital procurement affects innovation through two mechanisms. First, contract design may shape the direction of innovation through constraints on contractible solutions (Obwegeser and Müller, 2018; Uyarra et al., 2020). Project-based and highly customized contracts restrict the feasible innovation space, steering suppliers toward mature, contractible refinements and system-compatible solutions rather than high-risk, frontier experimentation (Georghiou et al., 2014; Howell et al., 2021). This specification-constrained paradigm is further reinforced by technical demands for security, reliability, and interoperability inherent in digital public goods. In practice, such bureaucratic complexity and rigid compliance requirements can generate

tem Government Procurement Demand Standard (2023 Edition) includes mandatory indicators on kernel compatibility, long-term compatibility support, and system recovery (`https://www.mof.gov.cn/jrttts/202312/t20231226_3924138.htm`).

substantial passive waste, as excessive regulatory burden makes procurement cumbersome and raises procurement costs (Bandiera et al., 2009).

As digital projects scale up, requirement setting increasingly relies on standardization tools, such as national or ministry-issued demand standards. While these standards offer standardized benchmarks to mitigate arbitrariness, standard-setting in volatile digital domains is inherently prone to temporal lags. When procurement requirements are anchored to slowly updated standards or narrowly defined catalogs, solution choices may become locked into inferior, incumbent-compatible options under incomplete information (Farrell and Saloner, 1985). Furthermore, to integrate with incumbent public systems, suppliers often face substantial interoperability challenges and switching costs (Bajari et al., 2014; Farrell and Klemperer, 2007). Because complex digital projects involve high ex-post adaptation costs, rigid contract specifications inevitably create massive bargaining frictions when technological adjustments are needed (Bajari and Tadelis, 2001). This structural rigidity can induce a lock-in effect, where public procurement rewards highly bespoke, project-specific engineering rather than broadly deployable technological innovation.

Second, non-market allocation forces in procurement distort both incentives and selection. When contract awards are influenced by local favoritism, selection can become less sensitive to technological merit and more dependent on geographic proximity and relationship-specific advantages (Tang et al., 2025). Accreditation-based policy instruments can also be implemented in ways that reinforce regional segmentation and protectionist screening (Li and Georghiou, 2016). As a result, firms may have stronger incentives to invest in local embeddedness, political access, and other non-productive activities rather than in broadly applicable innovative capabilities (Akcigit et al., 2023). More broadly, political connections can improve firm survival and revenue growth without generating commensurate productivity gains, and connected allocation can weaken real economic outcomes (Choi et al., 2024).

In sum, government digital procurement may diminish innovative economic efficiency by not only steering firms toward narrow, contract-specific adaptation but also by privileging political embeddedness over technological superiority, thus weakening the translation of R&D capital into tangible economic value. We therefore posit:

H1: Government digital procurement has a negative impact on firms' innovative economic efficiency.

3.2. The cost structure mechanism

Government digital procurement can shift firms' cost structure toward a more variable and asset-light form. The managerial accounting literature emphasizes that cost behavior is not a purely technical function; it reflects managerial choices under adjustment costs and incentive constraints (Banker et al., 2014; Dierynck et al., 2012). When demand is project-based and performance is tightly monitored, managers have incentives to avoid irreversible commitments and substitute fixed resources with adjustable inputs (outsourcing, flexible labor, leasing, pay-per-use infrastructure) as a form of risk management (Aboody et al., 2018; Holzhacker et al., 2015; Kallapur and Eldenburg, 2005). This substitution is particularly natural for digital products and services, which are inherently scalable. Compared with traditional lumpy capacity expansion, digital service provision can be ramped up or down more continuously by purchasing external modules, cloud capacity, or specialized labor in response to order volume. As a result, the redundant fixed resources that would otherwise be held to meet peak demand are partly replaced by costs that co-move with contract size, implying a decline in operating leverage and cost rigidity. Empirically, evidence on digital technology adoption also points to lower internal adjustment frictions and more flexible operations (Hui et al., 2024; Li et al., 2024), reinforcing the prediction that exposure to digital procurement shifts firms toward a more variable cost structure.

A more flexible cost structure may reduce innovative economic efficiency through two related channels. First, by lowering the fixed-cost share, it weakens operating leverage, so that revenue growth translates less strongly into profits (Aboody et al., 2018; Novy-Marx, 2011). In an innovation context, this matters because the value of R&D depends on whether new knowledge can be commercialized in a scalable way. When revenue growth must be supported by pay-per-unit inputs, successful innovation yields less retained surplus because each additional sale requires a larger contemporaneous variable-cost outlay. As a result, the firm captures a smaller share of the value created by innovation, so the revenue payoff to a given stock of accumulated R&D capital is weakened. In this sense, a more flexible cost structure does not necessarily reduce innovation input, but it can reduce the scalability of innovation returns and thereby lower innovative economic efficiency.

Second, an asset-light cost structure lowers exit and switching costs, which can undermine the organizational commitment needed for long-horizon innovation. With fewer quasi-fixed resources embedded in stable teams and complementary assets, firms can more easily downsize, pivot, or abandon projects when interim performance fluctuates. While such flexibility helps manage short-term risk, innovation incentives often require tolerance for early failure and persistence to realize long-run payoffs (Manso, 2011). Accordingly, the procurement-induced shift toward lower cost rigidity can weaken cumulative capability building and reduce the effectiveness with which R&D is transformed into revenue-generating capabilities, consistent with lower innovative economic efficiency. We therefore posit:

H2: Government digital procurement weakens firms' cost structure rigidity, which in turn lowers their innovation economic efficiency.

4. Data

4.1. Variable definitions

4.1.1. Identifying governmental digital procurement

To identify government digital procurement contracts awarded to listed firms, we first link government contracts to listed firm entities and then classify the matched contracts as digital procurement orders according to the *Statistical Classification of the Digital Economy and Its Core Industries (2021)*.

Accurate entity matching is the foundation for constructing the government digital procurement indicator. Because suppliers in government procurement projects may be subsidiaries rather than parent listed firms, we first compile a comprehensive name dictionary covering all listed firms and their subsidiaries. To improve match quality, we normalize both supplier names in the procurement data and firm names in the listed-firm registry by removing common organizational suffixes, such as "有限公司" (Co., Ltd.), "股份有限公司" (Co., Ltd.), "有限总公司", "有限分公司", "集团" (Group),"公司" (Company) and "企业" (Enterprise). We then use the term frequency–inverse document frequency (TF-IDF) method to convert the normalized names into numerical vectors and compute cosine similarity scores between supplier names and listed-firm entities. We adopt 0.6 as the initial matching threshold: observations with similarity above 0.7 are treated as exact matches[12], while those with scores between 0.6 and 0.7 are manually reviewed to ensure matching accuracy. Using this procedure, we match 1,853 listed firms to 93,270 government procurement contracts.

Given the matched contract sample, we next identify whether each contract falls within the scope of the digital economy. Rather than relying on a keyword dictionary (e.g., Chen et al., 2025), we use supervised semantic classification. This choice is motivated by the fact that the observable text fields are short—limited to the contract title and primary item—whereas the official scope of the digital economy is conceptually broad and difficult to capture with a small, stable set of keywords[13]. Dictionary-based meth-

[12]In selecting the similarity cutoff, we compared multiple threshold values. A cutoff of 0.7 provides a reasonable balance between false positives and false negatives.

[13]Table Appendix A.2 reports representative examples showing that seemingly relevant terms do not reliably identify digital procurement contracts, whereas Figure Appendix A.1 illustrates the information typically disclosed in raw procurement records, such as the contract title and primary item.

ods are therefore prone to both low recall and low precision in this setting (Gentzkow et al., 2019; Grimmer and Stewart, 2013). To address this issue, we fine-tune a Chinese BERT model with whole-word masking (BERT-wwm) for binary classification. Compared with prompt-based labeling by large language models, this approach is better suited to a measurement procedure that emphasizes reproducibility and consistency at scale (Cui et al., 2021; Devlin et al., 2019). The identification process involved four main steps:

First, a random sample of 5,000 records was drawn from the matched procurement contracts. Based on the criteria established in the Statistical Classification of the Digital Economy and Its Core Industries (2021) by the National Bureau of Statistics, we manually annotated the contract content and subject matter to construct a ground-truth dataset for binary classification(i.e., whether a project belongs to the digital economy).

Second, this annotated dataset was partitioned into training, validation, and test sets in an 8:1:1 ratio. The pre-trained Chinese BERT-wwm model was then fine-tuned on the training set—utilizing the validation set for hyperparameter tuning—to perform the specific text classification task of identifying digital economy procurement.

Third, the out-of-sample performance of the fine-tuned model was evaluated on the held-out test set. The model achieved an accuracy of 94%, demonstrating strong generalization capabilities. Subsequently, the optimized model was deployed to conduct inference on the entire matched government procurement dataset of listed firms, automatically identifying and labeling a total of 46,837 digital economy-related procurement contracts.

Finally, the identified digital economy procurement contracts were aggregated at the firm-year level (inclusive of affiliated companies). We define our core explanatory variable, $DigProc\_Value$, as the total annual value of digital economy contracts to each listed firm, measured in yuan. Ultimately, this aggregation yields a total of 3,824 firm-year observations with positive digital procurement values, covering 1089 listed firms over the sample period.

4.1.2. Measuring innovative economic efficiency

Our measure of RQ is constructed following Cooper et al. (2022). Specifically, under a Cobb–Douglas production technology, we define RQ for firm i as the output elasticity with respect to R&D capital, namely the exponent α_{1i} in the following production function:[14]

$$Y_{it} = A_{it} R_{i,t-1}^{\alpha_{1i}} K_{it}^{\alpha_{2i}} L_{it}^{\alpha_{3i}} S_{i,t-1}^{\alpha_{4i}} D_{it}^{\alpha_{5i}} e_{it}, \tag{1}$$

where Y_{it} denotes firm revenue, $R_{i,t-1}$ is the stock of R&D capital lagged by one year, K_{it} is the physical capital stock, L_{it} is the number of employees, $S_{i,t-1}$ captures knowledge spillovers from technologically more advanced peers, D_{it} denotes the stock of advertising proxied by selling expense, A_{it} is total factor productivity, and e_{it} is an idiosyncratic error term. Following Cooper et al. (2022), we define the peer knowledge spillover term as $S_{it} = \sum_j (R_{jt} - R_{it}) \mathbf{1}(R_{jt} \geq R_{it})$, where the summation is taken over firms in the same industry, and $\mathbf{1}(\cdot)$ is an indicator function equal to one if firm j has a higher level of R&D capital than firm i, and zero otherwise. This formulation captures the idea that firms can benefit more from knowledge generated by technologically leading peers, with the potential spillover increasing in the technological distance between the frontier firms and the focal firm. Both the R&D input and the spillover term are lagged by one period to allow for the time needed for knowledge accumulation and transformation into output.

The key difference in our construction is that we use stock rather than flow measures of the input variables R, S and D in the benchmark estimation. This choice is guided by the discussion in Cooper et al. (2022) about whether R&D should be modeled using stocks or flows. While they argue that, under a steady-state assumption, flow-based and stock-based specifications yield

[14] RQ has the advantage of directly reflecting a firm's ability to translate R&D inputs into economic returns. Relative to patent data, it is also less affected by truncation arising from confidentiality concerns and delays in patent examination, and is therefore better suited to measuring the economic efficiency of innovation.

very similar elasticities, they also acknowledge that stocks are theoretically closer to the underlying knowledge capital. In the rapidly evolving innovation environment of Chinese listed firms, we expect R&D stock, cumulative advertising investments and knowledge spillovers to influence current revenue.

To implement this stock-based approach, following the robustness design of Zhang and Yang (2025), we first deflate the annual flows of R&D expenditure and selling expenses by the producer purchase price index and the producer ex-factory price index, respectively.[15] We then construct real stocks by applying the perpetual inventory method. For input $I \in (R, D)$, the stock evolves according to $I_{it}^{\text{stock}} = (1 - \delta_I) I_{i,t-1}^{\text{stock}} + I_{it}^{\text{flow}}$, where δ_I is the depreciation rate.

We set the depreciation rate for R&D to 15% and for selling expenses to 67%, and initialize the corresponding stocks at zero in the first year they enter the sample.[16] The stock of knowledge spillovers, S_{it}, is constructed by aggregating the R&D stock gaps between firm i and its more innovative peers within the same industry. As fixed assets, K_{it}, are stock variables, we deflate them using the fixed-asset investment price index to ensure comparability over time.

To obtain firm-year specific estimates of RQ, we log-linearize the production function in Eq. (1):

$$\ln Y_{it} = \alpha_{0i} + \alpha_{1i} \ln R_{i,t-1} + \alpha_{2i} \ln K_{it} + \alpha_{3i} \ln L_{it} + \alpha_{4i} \ln S_{i,t-1} + \alpha_{5i} \ln D_{it} + \varepsilon_{it}, \tag{2}$$

where α_{1i} is the firm-specific elasticity of revenue with respect to R&D. We estimate Eq. (2) using a random-coefficient model, which allows the elastic-

[15] We use 2001 as the base year for the relevant price indices.

[16] In our data, firm-level R&D expenditure records are available from 2006 onward, and R&D capital is therefore accumulated recursively from the earliest available firm-level R&D observations. The 6-year rolling-window estimation of RQ for the year 2015 utilizes historical accounting data starting from 2010, ensuring that our initial RQ estimates are built on a sufficient accumulation of prior data.

ities of all inputs to vary across firms (Cooper et al., 2022). To introduce time variation in RQ, we apply the model over 6-year rolling windows, such that the estimate for year t is based on data from $[t-5, t]$. In each window, we retain firms with at least four valid R&D observations. After truncating the sample to the 2015–2023 to align with the availability of the government digital procurement panel, our final RQ sample consists of 16,189 firm-year observations across 2,985 firms.

4.1.3. Controls

To mitigate potential omitted variable bias, we incorporate a comprehensive set of control variables covering financial characteristics, market valuation, and corporate governance. These include Age (natural logarithm of years since listing), BM (book-to-market ratio), Cur (current assets scaled by total assets), Dual (dummy variable indicating whether the CEO also serves as the Chairman), Lev (total liabilities scaled by total assets), ROA (return on assets), Size (natural logarithm of total assets), SOE (a dummy for state-owned enterprises), Tobin's Q, and Top1 (the share of the largest shareholder). In all primary specifications, we further include firm, year, industry, and city fixed effects.

4.2. Data sources and sample selection

Because government procurement contract data are limited and incomplete prior to 2015, our empirical analysis uses detailed procurement records from 2015 to 2023. The raw contract data are collected from the official portal of the China Government Procurement Network, and firm-level financial and governance information is drawn from the CSMAR database. We construct the main panel by merging three data sources: listed-firm financial and governance data, the government digital procurement dataset, and firm-year-specific RQ estimates. For firm-years without identified digital procurement contracts, the procurement amount is set to zero. Following the literature, we exclude ST, *ST, and PT firms, remove firms in the financial industry, drop abnormal observations (e.g., negative revenues or negative net equity), and winsorize all continuous variables at the 1st and 99th percentiles within each year. Given the large magnitude of the raw contract amounts, we construct the final explanatory variable, $DigProc$, as the natural logarithm of one plus the procurement amount.

5. Results

5.1. Summary statistics

Table 1 presents the summary statistics for the variables used in the baseline analysis. The key explanatory variable, $DigProc$, is highly skewed, with a mean of 1.8883, a median of 0, and a maximum of 22.5255. This pattern indicates that government digital procurement is absent or limited for most firm-year observations, while a small subset of firms accounts for substantially larger procurement volumes. The dependent variable, RQ, has a mean of 0.1068. This indicates that a 1% increase in R&D investment generates a 0.1068% increase in revenue for a representative firm in our sample, which is slightly lower than that of the U.S. sample (0.141%) reported in Cooper et al. (2022). The control variables display reasonable dispersion overall, with no obvious abnormal patterns in their distributions.

[Table 1 about here.]

5.2. Baseline

To examine the effect of government digital procurement on firms' innovative economic efficiency, we construct the following baseline specification:

$$RQ_{it} = \beta_0 + \beta_1 DigProc_{it} + \Gamma \mathbf{X}_{it} + FE + \varepsilon_{it}, \tag{3}$$

where i and t index firm and year, respectively. The definitions of RQ_{it}, $DigProc_{it}$, and the vector of control variables $\mathbf{X}_{it}$ are provided in Section 4.1. FE denotes the fixed-effects structure adopted in each specification, including two-way firm and year fixed effects, as well as additional industry and city fixed effects. Standard errors are clustered at the firm level.

[Table 2 about here.]

Table 2 reports the baseline estimation results. Column (1) includes $DigProc_{it}$ together with firm and year fixed effects. Column (2) further adds the full set of firm-level controls. Column (3) additionally includes industry and city fixed effects. Across all three specifications, the coefficient on $DigProc_{it}$ is negative and statistically significant at the 1% level. Based on the comprehensive specification in Column (3), the estimated coefficient of -0.0056 indicates that a one-standard-deviation increase in government digital procurement is associated with a 1.01 percentage-point standard-deviation decline in innovative economic efficiency.[17] These results provide support for Hypothesis H1.

We complement the baseline evidence by replacing the dependent variable in Eq. (3) with alternative measures of firms' R&D investment. Table Appendix A.1 shows that government digital procurement is positively associated with R&D inputs. Combined with the baseline estimates, these results suggest that government digital procurement raises firms' innovation effort, but is less effective in translating such inputs into innovation-driven economic returns and competitiveness.

5.3. Mechanism

To empirically validate this mechanism in Hypothesis 2, we adopt a two-stage strategy. First, we examine whether government digital procurement significantly reduces cost rigidity. Second, we analyze whether the cost structure induced by such procurement exerts a detrimental effect on innovative

[17] To quantify the economic magnitude in a scale-robust way, we calculate the standard-deviation-scaled measure proposed by Mitton (2024). Specifically, the standard-deviation-scaled measure is defined as $E = |\hat{\beta}| \frac{s_x}{s_y}$, where $\hat{\beta}$ is the estimated coefficient on $DigProc_{it}$, s_x is the sample standard deviation of $DigProc_{it}$, and s_y is the sample standard deviation of the dependent variable. Since our regressions use $100 \times RQ$ as the dependent variable, $s_y = 100 \cdot \mathrm{sd}(RQ)$. Using the summary statistics reported in Table 1, where $s_x = 5.2156$ and $s_y = 2.88$, together with the estimated coefficient $\hat{\beta} = -0.0056$, we obtain $E \approx 0.0101$.

economic efficiency.

5.3.1. Government digital procurement and cost structure

Following Banker et al. (2014), to examine whether government digital procurement affects firms' cost structure, this paper estimates the following cost–sales relation:

$$\begin{aligned}\Delta \ln Cost_{it} =& \gamma_0 + \gamma_1 DigProc_{it} \times \Delta \ln Sale_{it} + \gamma_2 DigProc_{it} + \\ & \gamma_3 \Delta \ln Sale_{it} + \Xi \mathbf{Z}_{it} + FE + \xi_{it},\end{aligned} \tag{4}$$

where $\Delta \ln Cost_{it}$ denotes the change in the logarithm of operating costs of firm i in year t, and $\Delta \ln Sale_{it}$ is the corresponding change in the logarithm of sales revenue. Following Banker et al. (2014), the control vector Z_{it} comprises a set of firm characteristics: Age, Lev, $Size$, SOE, $Top1$, $Labor\_intensity$ and $Capital\_intensity$. The first five variables are defined as in Section 4.1, $Labor\_intensity$ is calculated as the number of employees divided by sales revenue, and $Capital\_intensity$ is measured as total assets scaled by sales revenue. In this framework, the slope of costs with respect to sales captures the degree of cost flexibility

$$\frac{\partial \Delta \ln Cost}{\partial \Delta \ln Sale} = \gamma_3 + \gamma_1 DigProc_{it}, \tag{5}$$

the coefficient γ_1 is the parameter of primary interest. A positive and significant γ_1 implies that, for firms with a higher level of government digital procurement, costs react more strongly to changes in sales, indicating a more flexible cost structure. Conversely, a negative γ_1 would suggest that digital procurement increases cost rigidity.

[Table 3 about here.]

Table 3 reports the estimation results for the above specification. In all columns, the estimated coefficient $\hat{\gamma}_1$ on $\Delta \ln Sale_{it} \times DigProc_{it}$ is positive and statistically significant at the 1% level. These findings indicate that, conditional on firm characteristics, a higher level of government digital procurement increases the sensitivity of operating costs to demand fluctuations and leads to a more flexible cost structure.

5.3.2. Procurement-induced cost flexibility and RQ

This subsection further investigate how procurement-induced variation in cost flexibility would affect RQ, following the logic of causal mediation analysis (Celli, 2022; Imai et al., 2010). Hanlon et al. (2015) estimate year-specific tax coefficients using separate regressions for each year. While intuitive, this approach is potentially inefficient because each estimation uses only within-year observations, and the resulting coefficients remain homogeneous across firms in the same period. Such a framework is therefore ill suited to our purpose of identifying cross-sectional heterogeneity in the effect of government digital procurement on cost flexibility.

To address this issue, we follow Zhang and Yang (2025) and adopt a random-coefficient model, which allows the cost sensitivity to sales associated with government digital procurement to vary across firms. Specifically, we estimate the following empirical specification:

$$\begin{aligned}\Delta \ln Cost_{it} =& \gamma_0 + \gamma_{1i} DigProc_{it} \times \Delta \ln Sale_{it} + \gamma_2 DigProc_{it} + \\ & \gamma_3 \Delta \ln Sale_{it} + \Xi \mathbf{Z}_{it} + FE + \xi_{it}.\end{aligned} \tag{6}$$

All the variables are the same as Eq. (4). Based on the estimated firm-specific slopes $\hat{\gamma}_{1i}$, we construct a firm-year index, $M_{it} = \hat{\gamma}_{1i} \times DigProc_{it}$, to capture the extent of cost-structure change induced by government digital procurement. A larger value of M_{it} indicates a more flexible cost structure induced by government digital procurement. We then examine how this index is related to firms' innovative economic efficiency by estimating the following regression:

$$RQ_{it} = \theta_0 + \theta_1 M_{it} + \Theta \mathbf{X}_{it} + FE + \zeta_{it}, \tag{7}$$

where X_{it} is the set of control variables used in the baseline regressions. The coefficient θ_1 reflects how digital-procurement-induced changes in cost structure are associated with firms' innovative economic efficiency.

[Table 4 about here.]

Table 4 reports the estimation results for Eq. (7). Across all three specifications, the coefficient on M_{it} is negative and highly significant at the 1% level. This evidence indicates that firms subject to a larger procurement-induced increase in cost flexibility systematically exhibit lower innovative economic efficiency. The result supports Hypothesis H2 that government digital procurement affects firms' innovative performance through the cost-structure channel.

6. Robustness

To verify that our findings are not sensitive to specific modeling choices or sample construction, we conduct a series of robustness checks.

6.1. Controlling for high-dimensional fixed effects

We re-estimate the core regressions controlling for high-dimensional fixed effects. We augment our baseline model with Industry-Year and City-Year fixed effects, which sequentially absorb time-varying industry characteristics and regional macroeconomic shocks. Furthermore, we introduce City-Firm coupled with Year fixed effects to meticulously account for location-specific firm heterogeneity.

[Table 5 about here.]

In Table 5, Panel A displays the results for the main effect of government digital procurement on innovative economic efficiency (RQ_{it}). Panel B and Panel C verify the robustness of the mechanism. Across all three columns, the empirical results exhibit a high degree of consistency with our previous findings.

6.2. Instrumental variable approach

To address the potential endogeneity of government digital procurement $DigProc_{it}$ in the baseline and mechanism regressions, we re-estimate Eq (3) (4) and (7) using three sets of instruments.

Industry-average procurement. Following Fisman and Svensson (2007) , we construct the average government digital procurement of other firms in the same industry and year as our instrument. This variable is strongly correlated with firm-specific procurement intensity but is plausibly exogenous, because it captures common shocks at the industry level rather than firm-specific demand or innovation shocks.

Bartik's instrumental variable. To exploit exogenous demand shocks, we construct a Bartik instrument (Bartik, 1991; Goldsmith-Pinkham et al., 2020) that interacts initial local shares with national growth rates. The formula is specified as:

$$Bartik_{pt} = \sum_{j=1}^{J} \frac{DigProc_{p,j,0}}{DigProc_{p,0}} \times GrowNationDigProc_{-p,-j,t} \tag{8}$$

where $\frac{DigProc_{p,j,0}}{DigProc_{p,0}}$ is the base-year share of government digital procurement orders in industry j within province p, and $GrowNationDigProc_{-p,-j,t}$ is the growth rate of government digital procurement for industry j in all other provinces and industries (excluding province p and industry j) at time t. This shift–share term captures how strongly province p is exposed to nation-wide digital procurement shocks, while purging local policy or firm-level innovation decisions.

Lewbel's instrumental variable. We adopt the approach of Lewbel (2012), which generates internal instruments using the heteroskedasticity of first-stage residuals. Specifically, if $Cov(Z, \epsilon^2) \neq 0$ and $Cov(Z, \epsilon) = 0$, then $(Z - \bar{Z})\widehat{\epsilon}$ can serve as valid instruments. In implementation, we use the annual industrial means of the control variables vector as the auxiliary vector Z. These variables are plausibly exogenous because they capture broad industry-year conditions rather than firm-specific procurement demand or innovation shocks. Moreover, in the Lewbel specification, we replace separate industry and year fixed effects with industry-year interacted fixed effects, while retaining firm and city fixed effects, so as to absorb time-varying industry-wide shocks more tightly and reduce the concern that the identifying variation is driven by omitted industry-year confounders.

[Table 6 about here.]

For the baseline model (Panel A) and the second step of the mechanism (Panel C), we employ standard 2SLS. For the first step of the mechanism (Panel B), which involves an endogenous interaction term, we adopt a two-stage substitution procedure (Gulen and Ion, 2016): (*i*) regress the endogenous variable $DigProc_{it}$ on the instrument IV_{it} and controls to obtain the fitted value $\widehat{DigProc}_{it}$; (*ii*) substitute the fitted value $\widehat{DigProc}_{it}$ directly into the interaction term of the cost structure equation Eq. (4).

Table 6 presents the IV estimation results. The first-stage Kleibergen–Paap Wald F-statistics for IV_Ind_avg (255.98), IV_Bartik (12.93), and IV_Lewbel (349.82) all exceed conventional critical thresholds, rejecting weak identification. For Lewbel's IV, the Breusch–Pagan statistic (829.65) confirms the heteroskedasticity condition required for identification. Overall, the IV estimates continue to support a negative effect of government digital procurement on firms' innovative economic efficiency. The mechanism estimates are also broadly consistent with the cost-structure channel.

6.3. Alternative clustering of standard errors

Considering potential regional spillover effects of government digital procurement policies or common macroeconomic shocks inducing correlations among firms in specific years, clustering solely at the firm level might underestimate the standard errors. To alleviate this concern, and following the recommendations of Abadie et al. (2023), we conduct robustness checks using alternative clustering specifications. To address broader spatial, sectoral, and temporal correlations, we re-estimate the baseline and mechanism models by clustering standard errors at the city level, the industry level, and the firm-year two-way levels. As shown in Table 7, our core findings remain statistically significant under these alternative clustering specifications.

[Table 7 about here.]

6.4. Alternative measures of core variables

6.4.1. Alternative measures of DigProc

To mitigate the potential endogeneity concern arising from reverse causality, whereby a firm's current innovative efficiency may affect its acquisition of digital contracts, we re-estimate the models using one- and two-period lagged values of government digital procurement. Additionally, to ensure our findings are not driven by variable scaling, we employ a standardized measure of digital procurement.

[Table 8 about here.]

As reported in Table 8, both the baseline negative impact on *RQ* (Panel A) and the underlying cost-structure mechanism (Panels B and C) remain robust under these alternative specifications.

6.4.2. Alternative constructions of RQ

Our baseline measure follows Cooper et al. (2022) and constructs *RQ* using a six-year rolling window. To assess the sensitivity of our findings to this choice, we re-estimate Eqs. (3) and (7) using alternative rolling windows of four and eight years. We do not repeat Eq. (4), because it does not rely on the estimation of *RQ*. As shown in Table 9, the negative effect of digital procurement is preserved across these alternative window lengths.

We also consider whether our findings depend on using stock rather than flow measures in constructing *RQ*. The benchmark specification adopts stock-based inputs to reflect the cumulative effect of past investment, while this robustness check follows Cooper et al. (2022) and re-estimates *RQ* based on flow measures. The results in Table 9 remain qualitatively similar under this alternative measure. Finally, replacing *RQ* with the natural logarithm of revenue-to-lagged-R&D-stock ratio, $\ln(Y_{i,t}/K_{i,t-1})$, leads to similar conclusions.

[Table 9 about here.]

7. Heterogeneity analysis

We next identify the organizational and institutional conditions under which the negative effect of government digital procurement on innovative economic efficiency is either amplified or mitigated. The empirical model is specified as follows:

$$RQ_{it} = \eta_1 DigProc_{it} + \eta_2 \big(DigProc_{it} \times H_{it} \big) + \eta_3 H_{it} + \Gamma \mathbf{X}_{it} + FE + \varepsilon_{it}, \quad (9)$$

where H_{it} is a binary indicator that alternatively captures state ownership, initial asset rigidity, industry competition, repeated supplier status, and province-level market institutions. For transparency, we include both the interaction term and the main effect of H_{it}. When H_{it} is absorbed by fixed effects, the coefficient on H_{it} is not separately identified; nevertheless, the coefficient on the interaction term $DigProc_{it} \times H_{it}$ remains identified. The coefficient of interest is η_2, which measures the differential effect of government digital procurement between the two groups.

[Table 10 about here.]

(a) State ownership (SOE). Ownership type is a key feature of Chinese listed firms. We define SOE_i as an indicator equal to one if firm i is ultimately controlled by the central government, local governments, or other state entities, and zero otherwise. Column (1) of Table10 shows that the interaction term $DigProc_{it} \times SOE_i$ is negative and statistically significant, implying that the marginal effect of $DigProc_{it}$ for SOEs is more negative than that for non-SOEs. This pattern is consistent with the view that SOEs are more state-embedded and operate under multiple objectives, which may place greater weight on compliance, delivery, and policy considerations than on purely commercial value creation (DeWenter and Malatesta, 2001; Faccio, 2006; Shleifer and Vishny, 1994). Taken together, the SOE heterogeneity suggests that the innovation consequences of digital procurement are shaped by ownership-based incentive structures, amplifying the decline in RQ when firms face stronger state-linked objectives.

(b) Long-term suppliers. We define long-term suppliers as firms that win government digital procurement contracts at least twice during 2015–2023, and this indicator is time-invariant at the firm level. Column (2) of Table 10 indicates that the negative effect of digital procurement is stronger for firms with repeated wins in the sample period. A plausible interpretation is that repeated procurement relationships in digital projects are more likely to involve relationship-specific investments and compatibility constraints, raising switching costs and strengthening incumbency advantages (Baker et al., 2002; Klemperer, 1987). Such lock-in dynamics can encourage continued effort toward contract-specific adaptation and system maintenance, with weaker incentives to codify and diffuse innovations into scalable market offerings, thereby depressing the revenue productivity of R&D capital.

(c) Initial cost rigidity. Initial cost rigidity is proxied by the firm's fixed-asset intensity, measured as the average ratio of fixed assets to total assets over 2015–2016. Column (3) of Table10 shows that the negative effect of government digital procurement on RQ is substantially attenuated for firms with higher initial cost rigidity.[18] This pattern is consistent with the view that firms with more rigid cost structures face higher adjustment costs and are less likely to shift toward highly flexible, contract-specific resourcing in response to digital procurement (Cooper and Haltiwanger, 2006). As fixed-asset intensity captures a longer-horizon cost-structure choice, it may shape whether procurement-driven demand is absorbed as one-off delivery or converted into more scalable capabilities (Balakrishnan et al., 2014; Banker et al., 2014). Since the rigidity measure is based on an early-period average and held fixed over time, it is more naturally interpreted as a predetermined organizational characteristic.

(d) Industry competition. Industry concentration is measured by the lagged Herfindahl–Hirschman Index (HHI) constructed from firms' operat-

[18]We use the two-year average rather than the 2015 value alone to reduce the influence of missing observations in a single year. This firm-level measure is held constant over the sample period and split at the sample median to form a high-rigidity indicator.

ing revenue shares within each industry-year. For each year, we perform an equal-sized split based on HHI_{jt-1} and define $high\_comp = 1$ for the half of industries with lower HHI. Column (4) of Table10 suggests that stronger product-market competition mitigates the adverse impact of government digital procurement on innovative economic efficiency. A natural interpretation is that intense competition strengthens market discipline and compresses slack, forcing firms to convert project-based demand into commercially viable and scalable offerings, rather than allocating innovation effort primarily toward bespoke delivery and compatibility work. This is consistent with evidence that competition improves corporate performance by sharpening incentives and limiting managerial slack (Nickell, 1996). Moreover, competition can raise the returns to innovation when firms seek to differentiate and "escape" rivals, which may increase the likelihood that procurement-driven innovation is codified and redeployable beyond a single contract (Aghion et al., 2005). Overall, the competition heterogeneity points to a key boundary condition: the same digital procurement shock is less detrimental to RQ when firms operate under stronger product-market pressure.

(e) Intermediaries and legal environment. The measure is based on the NERI (Fan Gang) provincial marketization index sub-dimension commonly referred to as the *development of market intermediaries and the legal environment*. For each year, we compute the annual median of this sub-index across provinces and define a high institutional-quality indicator for provinces at or above the median. Column (6) of Table10 shows that the negative impact of government digital procurement on RQ is weaker in provinces with a more developed intermediaries-and-legal environment.[19] A plausible interpretation is that stronger contract enforcement and institutional infrastructure reduce opportunistic frictions and strengthen firms' incentives to undertake longer-horizon, redeployable innovation investments. Consistent

[19]The original series is available through 2019; for 2020–2023 we extrapolate province-specific values using the average annual growth rate calculated from the available historical series.

with this view, cross-country evidence links better contract enforcement to higher R&D intensity, especially in contract-dependent settings (Seitz and Watzinger, 2017), and judicial reforms that improve court effectiveness are associated with meaningful productivity gains (Chemin, 2020). Moreover, when legal and intermediary services are more developed, firms may find it easier to scale and commercialize procurement-induced innovations, consistent with the broader profiting-from-innovation perspective (Teece, 1986). Overall, the institutional heterogeneity suggests that the same digital procurement shock is less likely to be absorbed as one-off, compliance-oriented delivery when the provincial legal–intermediary environment better supports credible contracting and market-based commercialization.

8. Conclusion

This paper examines how government procurement of digital products and services affects the innovative economic efficiency of Chinese listed firms. We identify digital procurement contracts from Chinese government procurement data using BERT-wwm text classification and estimate innovative economic efficiency as the revenue elasticity of R&D through a rolling random-coefficient model. We find that government digital procurement increases observable R&D input, but undermines firms' innovative economic efficiency by weakening cost structure rigidity. This adverse effect is concentrated among state-owned enterprises and repeat contract winners, while being attenuated for firms with higher initial cost rigidity, stronger product-market competition, and better provincial legal and intermediary institutions.

These findings do not suggest that government digital procurement should be scaled back. The coexistence of higher R&D input and lower innovative economic efficiency provides firm-level evidence for recent procurement reforms aimed at promoting fair competition, strengthening demand management, and reducing implementation distortions. For example, the Ministry of Finance's 2019 notice on fair competition targeted discriminatory clauses, ownership-based restrictions, local barriers, and other practices that undermine equal supplier participation.[20] The Measures for Government Procurement Demand Management further require procurement needs to be defined in a scientific, reasonable, and performance-oriented manner, with demand investigation and risk control for complex projects.[21] More recently, the State Council General Office's three-year action plan for 2024–2026 has elevated market-order rectification, legal-system construction, and innovation-oriented procurement into an integrated reform agenda.[22] Our results suggest that these reforms serve as substantive safeguards for steering digital procurement toward high-quality innovation, particularly when supported by competitive supplier selection, transparent disclosure, and credible contract governance.

The second policy implication is that digital procurement should be assessed less by detailed technical compliance or contract completion, and more by whether it generates functional, interoperable, and reusable solutions. For complex digital projects, procurement design should avoid excessive customization and encourage suppliers to convert public-sector demand into technologies that can be redeployed across broader markets. Recent reforms have begun to move in this direction. The 2021 demand-management measures require procurement needs to reflect project objectives and performance targets, while the 2024 Interim Measures for Cooperative Innovation Procurement introduce a "co-development plus first procurement" mechanism that emphasizes R&D targets, application scenarios, iterative upgrading, and the purchase of successful innovation products.[23] In this sense, digital procurement should not only solve project-specific problems, but also help suppliers transform customized delivery experience into reusable and commercially scalable capabilities.

[20] `https://m.mof.gov.cn/czxw/201907/t20190730_3319089.htm`.

[21] `https://www.gov.cn/gongbao/content/2021/content_5623061.htm`.

[22] `https://www.gov.cn/gongbao/2024/issue_11466/202407/content_6963171.html`.

[23] `https://gks.mof.gov.cn/guizhangzhidu/202404/t20240426_3933550.htm`.

Finally, digital procurement supervision should pay closer attention to suppliers that are more likely to face non-commercial objectives or relationship-specific lock-in. For state-owned suppliers and repeat contract winners, procurement authorities should strengthen post-award assessment and examine whether repeated awards or compatibility requirements create switching barriers. For complex digital projects, evaluation should move beyond timely delivery and technical compliance to consider whether suppliers generate reusable modules or transferable technological capabilities. In this regard, the innovation value of government digital procurement depends less on the scale of public demand than on whether procurement governance helps project-specific solutions evolve into scalable and redeployable capabilities.

Table 1: Summary statistics

Variables	Mean	Std.Dev	Min	Median	Max	Observations
Main variables						
DigProc	1.8883	5.2156	0	0	22.5255	16189
RQ	0.1068	0.0288	0.0149	0.1083	0.1984	16189
Controls						
Age	2.4614	0.5475	1.3863	2.4849	3.4340	16189
BM	0.6142	0.2637	0.0848	0.5972	1.2825	16002
Cur	0.5611	0.1835	0.0838	0.5729	0.9645	16189
Dual	0.2762	0.4471	0	0	1	15731
Lev	0.4157	0.1845	0.0523	0.4114	0.8627	16189
ROA	0.0493	0.0411	0.0011	0.0390	0.2400	16189
Size	22.5427	1.2361	19.9267	22.3762	26.6481	16189
SOE	0.3632	0.4809	0	0	1	16130
Tobin's Q	2.1013	1.3602	0.7797	1.6744	11.7922	16002
Top1	32.2673	14.2550	8.1816	30.0000	74.6689	16189

Notes: This table reports the summary statistics of the variables used in the baseline regressions. Control variables are defined in Section 4.1.3. The number of observations varies across variables because of missing values in some controls. For ease of coefficient interpretation, *RQ* is multiplied by 100 in the subsequent regressions.

Table 2: Government digital procurement and RQ

Dep. Var.: RQ_{it}	(1)	(2)	(3)
$DigProc_{it}$	-0.0046***	-0.0054***	-0.0056***
	(0.0011)	(0.0010)	(0.0010)
Controls		✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
Industry FE			✓
City FE			✓
Adjusted R^2	0.989	0.990	0.990
Observations	15895	15196	15194

Notes:This table reports the baseline regression results of firm innovative economic efficiency on government digital procurement. Standard errors, clustered at the firm level, are reported in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level.

Table 3: Government digital procurement and cost structure

Dep. Var.: $\Delta \ln Cost_{it}$	(1)	(2)	(3)
$\Delta \ln Sale_{it} \times DigProc_{it}$	0.0029*	0.0029**	0.0028**
	(0.0015)	(0.0014)	(0.0014)
Controls		✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
Industry FE			✓
City FE			✓
Adjusted R^2	0.879	0.879	0.876
Observations	14506	14451	14449

Notes: This table reports the first-step mechanism results on government digital procurement and firms' cost structure. The estimates are obtained by regressing the first difference of the logarithm of operating costs on the first difference of the logarithm of sales, the interaction term $\Delta \ln Sale_{it} \times DigProc_{it}$, and other control variables. The coefficient on the interaction term reflects the effect of government digital procurement on cost elasticity. Standard errors, clustered at the firm level, are reported in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level.

Table 4: Government digital procurement and cost structure

Dep. Var.: RQ_{it}	(1)	(2)	(3)
M_{it}	-0.0030***	-0.0034***	-0.0036***
	(0.0008)	(0.0007)	(0.0007)
Controls		✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
Industry FE			✓
City FE			✓
Adjusted R^2	0.988	0.990	0.990
Observations	13210	12707	12707

Notes: This table reports the second-step mechanism results linking procurement-induced changes in cost structure to firms' innovative economic efficiency. The dependent variable is RQ_{it}. The variable M_{it} measures the change in cost elasticity associated with government digital procurement. Standard errors clustered at the firm level are presented in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level.

Table 5: Controlling for high-dimensional fixed effects

	(1)	(2)	(3)
Panel A		RQ_{it}	
$DigProc_{it}$	-0.0041***	-0.0043***	-0.0054***
	(0.0009)	(0.0011)	(0.0010)
Observations	15136	14138	15196
Panel B		$\Delta \ln Cost_{it}$	
$\Delta \ln Sale_{it} \times DigProc_{it}$	0.0024*	0.0022†	0.0029**
	(0.0015)	(0.0015)	(0.0014)
Observations	14401	13510	14451
Panel C		RQ_{it}	
M_{it}	-0.0026***	-0.0026***	-0.0034***
	(0.0007)	(0.0008)	(0.0007)
Observations	12636	11747	12707
Controls	✓	✓	✓
Firm FE	✓	✓	
Year FE			✓
Industry-Year FE	✓		
City-Year FE		✓	
City-Firm FE			✓

Notes: This table presents the robustness of the previous findings under additional high-dimensional fixed effects. Panel A reports the baseline results for RQ_{it}, Panel B reports the first-step mechanism results for $\Delta \ln Cost_{it}$, and Panel C reports the second-step mechanism results for RQ_{it}. Column (1) introduces industry-year fixed effects, Column (2) introduces city-year fixed effects, and Column (3) introduces city-firm fixed effects together with year fixed effects. Standard errors clustered at the firm level are presented in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively. † indicates significance at the 10% level based on a one-sided (right-tailed) test.

Table 6: IV estimations

	IV_Ind_avg	IV_Bartik	IV_Lewbel
Panel A		RQ_{it}	
$DigProc_{it}$	-0.0335***	-0.0509*	-0.0077***
	(0.0125)	(0.0276)	(0.0027)
Kleibergen-Paap Wald statistic	255.980***	12.929***	349.820***
Breusch-Pagan statistic	–	–	829.65***
Observations	15194	13096	15194
Panel B		$\Delta \ln Cost_{it}$	
$\Delta \ln Sale_{it} \times DigProc_{it}$	0.0041***	0.0735***	0.0054***
	(0.0016)	(0.0021)	(0.0015)
Kleibergen–Paap Wald statistic	265.657***	12.587***	369.0***
Breusch–Pagan statistic	–	–	544.24***
Observations	14451	12036	14445
Panel C		RQ_{it}	
M_{it}	-0.0289***	-0.0402*	-0.0035
	(0.0107)	(0.0238)	(0.0022)
Kleibergen–Paap Wald statistic	192.097***	8.778***	309.503***
Breusch–Pagan statistic	–	–	640.21***
Observations	12707	10971	12707
Controls	✓	✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	
City FE	✓	✓	✓
Industry FE	✓	✓	
Industry-Year FE			✓

Notes: This table presents IV estimates using three alternative instruments for government digital procurement: the industry-average instrument, the Bartik instrument, and Lewbel's heteroskedasticity-based instrument. For Panels A and C, we re-estimate Eqs. (3) and (7) using two-stage least squares. For Panel B, we first obtain the fitted value of government digital procurement from the first-stage regression, and then substitute this fitted value into Eq. (4) to obtain the estimated results. Kleibergen–Paap Wald statistics test for weak instruments, and Breusch–Pagan statistics are additionally reported for Lewbel's IV. The Lewbel specification includes firm, city, and industry-year fixed effects. Standard errors are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 7: Robustness to alternative clustering of standard errors

	(1)	(2)	(3)
Panel A		RQ_{it}	
$DigProc_{it}$	-0.0056***	-0.0056***	-0.0056**
	(0.0011)	(0.0011)	(0.0017)
Observations	15194	15194	15194
Panel B		$\Delta \ln Cost_{it}$	
$\Delta \ln Sale_{it} \times DigProc_{it}$	0.0028**	0.0028	0.0028**
	(0.0014)	(0.0019)	(0.0012)
Observations	14449	14449	14449
Panel C		RQ_{it}	
M_{it}	-0.0036***	-0.0036***	-0.0036**
	(0.0008)	(0.0009)	(0.0012)
Observations	12707	12707	12707
Std. errors clustered by	City	Industry	Firm & year
Controls	✓	✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
City FE	✓	✓	✓
Industry FE	✓	✓	✓

Notes: This table reports the robustness of the baseline and mechanism results to alternative clustering schemes for standard errors. Column (1) clusters standard errors at the city level, Column (2) clusters at the industry level, and Column (3) applies two-way clustering by firm and year. Standard errors under each clustering scheme are reported in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level.

Table 8: Robustness: Alternative measures of *DigProc*

	Lagged		Standardized
	(1)	(2)	(3)
Panel A		RQ_{it}	
$DigProc_{it}$	-0.0043***	-0.0042***	-0.0111**
	(0.0011)	(0.0013)	(0.0054)
Observations	13940	12406	15194
Panel B		$\Delta \ln Cost_{it}$	
$\Delta \ln Sale_{it} \times DigProc_{it}$	0.0017	0.0030*	0.0100†
	(0.0017)	(0.0016)	(0.0075)
Observations	14449	12804	14449
Panel C		RQ_{it}	
M_{it}	-0.0028***	-0.0027***	-0.0073***
	(0.0006)	(0.0006)	(0.0027)
Observations	11773	10637	12707
Controls	✓	✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
City FE	✓	✓	✓
Industry FE	✓	✓	✓

Notes: This table presents the robustness of the previous findings using alternative measures of government digital procurement. Columns (1) and (2) use one-period and two-period lagged values of $DigProc_{it}$, respectively, while Column (3) uses a standardized measure of government digital procurement. Panel A reports the baseline results for RQ_{it}, Panel B reports the first-step mechanism results for $\Delta \ln Cost_{it}$, and Panel C reports the second-step mechanism results for RQ_{it}. Standard errors clustered at the firm level are presented in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level. † indicates significance at the 10% level based on a one-sided (right-tailed) test.

Table 9: Robustness: alternative measures of RQ

	RQ_{win4}		RQ_{win8}		RQ_{flow}		$ln(\mathrm{Y}_{\text{i,t}}/K_{\text{i,t-1}})$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$DigProc_{it}$	-0.0046***		-0.0044***		-0.0049***		-0.0033***	
	(0.0011)		(0.0011)		(0.0011)		(0.0012)	
M_{it}		-0.0028***		-0.0028***		-0.0032***		-0.0022**
		(0.0008)		(0.0008)		(0.0008)		(0.0009)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓	✓	✓
City FE	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This table presents the robustness of the baseline and second-step mechanism results under alternative measures of innovative economic efficiency. Columns (1)–(2) use a four-year rolling-window measure of RQ (RQ_{win4}), Columns (3)–(4) use an eight-year rolling-window measure (RQ_{win8}), and Columns (5)–(6) use a flow-based measure (RQ_{flow}). Columns (7)–(8) report results using $\ln(Y_{i,t}/K_{i,t-1})$ as a complementary revenue-based outcome measure. Standard errors clustered at the firm level are presented in parentheses.*, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 10: Heterogeneity analysis

Dep. Var.: RQ_{it}	(1) SOE	(2) Supplier	(3) Initial rigidity	(4) Ind-competition	(5) Legal-Intermediary
$DigProc_{it}$	-0.0030**	-0.0029*	-0.0084***	-0.0091***	-0.0098***
	(0.0013)	(0.0016)	(0.0015)	(0.0020)	(0.0019)
$DigProc_{it} \times SOE_i$	-0.0062***				
	(0.0020)				
$DigProc_{it} \times LongSupplier_{it}$		-0.0038*			
		(0.0020)			
$DigProc_{it} \times HighRig_i$			0.0060***		
			(0.0023)		
$DigProc_{it} \times HighComp_{it-1}$				0.0055**	
				(0.0023)	
$DigProc_{it} \times HighEnv_{it}$					0.0048**
					(0.0020)
Controls	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓
City FE	✓	✓	✓	✓	✓
Adjusted R^2	0.989	0.989	0.990	0.989	0.989
Observations	15196	15196	11075	15193	15186

Notes: This table reports the heterogeneity analysis for the effect of government digital procurement on firms' innovative economic efficiency. Columns (1)–(5) examine heterogeneity by state ownership, repeated supplier status, initial cost rigidity, industry competition, and the provincial legal–intermediary environment, respectively. In each column, the coefficient on $DigProc_{it}$ captures the effect of government digital procurement for the baseline group ($H_{it} = 0$), while the coefficient on the interaction term $DigProc_{it} \times H_{it}$ measures the differential effect for the corresponding high group ($H_{it} = 1$). Standard errors clustered at the firm level are presented in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level.

Appendix A. Additional Tables and Figures

Table Appendix A.1: Government digital procurement and R&D

Dep. Var.:	ln(*R&D*)	*R&D/Revenue*	*R&D* (Std.)
$DigProc_{it}$	0.0033**	0.0002***	0.0093***
	(0.0016)	(0.0001)	(0.0030)
Controls	✓	✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
Industry FE	✓	✓	✓
City FE	✓	✓	✓
Adjusted R^2	0.861	0.856	0.797
Observations	23782	23782	23782

Notes: This table reports complementary evidence on the relation between government digital procurement and firms' observable R&D inputs. The dependent variables are the log of R&D expenditure in Column (1), R&D intensity measured as R&D expenditure scaled by operating revenue in Column (2), and a standardized R&D measure in Column (3). All specifications include the full set of firm-level controls as well as firm, year, industry, and city fixed effects. Standard errors, clustered at the firm level, are reported in parentheses. *, **, *** correspond to significance at the 10%, 5%, and 1% level.

Table Appendix A.2: Justification for the Use of the BERT Model

Contract Name	Primary Item	Keywords	Digital Economy[a]
National Ambient Air Quality Monitoring Network Urban Air Automatic Monitoring Station Operation and Maintenance Project—O&M Service Entrustment Contract 2024 (Package 13)	National Ambient Air Quality Monitoring Network Urban Air Automatic Monitoring Station O&M (Package 13)	Automatic, Monitoring, Operation and Maintenance, O&M	Yes
Jiangsu Provincial People's Hospital Disinfection and Cleaning Products Framework Agreement Procurement Project Sterilization Powder (for WG-NQX-SR Fully Automatic Endoscope Washer-Disinfector) 001 Contract	Sterilization Powder	Fully Automatic	No
Zhangzhou City 2023 Natural Resources Survey and Monitoring Project	Zhangzhou City 2023 Forest, Grassland and Wetland Survey and Monitoring Municipal Pre-inspection	Monitoring	No
2023 Environmental Monitoring Outsourcing Project	Diesel Vehicle Exhaust Emission Testing Project	Monitoring	No
Changchun Municipal Facilities Maintenance and Management Center 2023 Municipal Infrastructure Maintenance and Operation Project (Urban Road Safety Operation, Road Disaster, Disease Prevention Testing and Pipeline Testing Assessment) Contract	2023 Municipal Infrastructure Maintenance and Operation Project (Urban Road Safety Operation, Road Disaster, Disease Prevention Testing and Pipeline Testing Assessment)	Testing, Maintenance	No
"Haixun 11" Vessel Operation and Maintenance Service Agreement—2024	Operation and Maintenance Service	Operation and Maintenance	No

Notes: This table presents representative examples used to illustrate why keyword-based identification is insufficient for classifying government digital procurement contracts and why the BERT model is needed. All original procurement contract texts are in Chinese; English translations are provided by the author. Although some contracts contain terms such as "Automatic," "Monitoring," or "Operation and Maintenance," these expressions do not necessarily indicate digital economy attributes. Accurate classification therefore requires semantic understanding based on the full contract context rather than simple keyword matching.
[a] Indicates whether the contract is classified as digital economy procurement (Yes/No).

Figure Appendix A.1
An Illustration of a Government Digital Procurement Contract

莆田市信息中心莆田市大数据调度指挥平台项目

2023-06-30 公告来源：中国政府采购网 【打印】

一、合同编号：[350301]PTHS[GK]2023003

二、合同名称：莆田市信息中心莆田市大数据调度指挥平台项目

三、项目编号：[350301]PTHS[GK]2023003

四、项目名称：莆田市信息中心莆田市大数据调度指挥平台项目

五、合同主体

采购人（甲方）：莆田市信息中心

地 址：莆田市城厢区市政府南广场5号楼

联系方式：13799006687

供应商（乙方）：福建省星云大数据应用服务有限公司,福建省大数据集团莆田有限公司

地 址：无

联系方式：17606063829

六、合同主要信息

主要标的名称：莆田市大数据调度指挥平台项目

规格型号（或服务要求）：详见合同

主要标的数量：1.0000项

主要标的单价：34980000.000000元

合同金额：3498.000000万元

履约期限、地点等简要信息：/

采购方式：公开招标

七、合同签订日期：2023-06-30

八、合同公告日期：2023-06-30

九、其他补充事宜：

附件：

免责声明：本页面提供的政府采购合同是按照《中华人民共和国政府采购法实施条例》的要求由采购人发布的，中国政府采购网对其内容概不负责，亦不承担任何法律责任。

References

Abadie, A., Athey, S., Imbens, G.W., Wooldridge, J.M., 2023. When should you adjust standard errors for clustering? The Quarterly Journal of Economics 138, 1–35. doi:10.1093/qje/qjac038.

Aboody, D., Levi, S., Weiss, D., 2018. Managerial incentives, options, and cost-structure choices. Review of Accounting Studies 23, 422–451. doi:10.1007/s11142-017-9432-0.

Aghion, P., Bloom, N., Blundell, R., Griffith, R., Howitt, P., 2005. Competition and innovation: An inverted-U relationship. The Quarterly Journal of Economics 120, 701–728. doi:10.1093/qje/120.2.701.

Akcigit, U., Baslandze, S., Lotti, F., 2023. Connecting to power: Political connections, innovation, and firm dynamics. Econometrica 91, 529–564. doi:10.3982/ECTA18338.

Arora, A., Belenzon, S., Cioaca, L.C., Ferracuti, E., 2025. The Private Value of Innovating for the Government. NBER Working Paper 33880. National Bureau of Economic Research. Cambridge, MA. doi:10.3386/w33880.

Aschhoff, B., Sofka, W., 2009. Innovation on demand—can public procurement drive market success of innovations? Research Policy 38, 1235–1247. doi:10.1016/j.respol.2009.06.011.

Bajari, P., Houghton, S., Tadelis, S., 2014. Bidding for incomplete contracts: An empirical analysis of adaptation costs. American Economic Review 104, 1288–1319. doi:10.1257/aer.104.4.1288.

Bajari, P., Tadelis, S., 2001. Incentives versus transaction costs: A theory of procurement contracts. Rand Journal of Economics 32, 387–407. doi:10.2307/2696361.

Baker, G., Gibbons, R., Murphy, K.J., 2002. Relational contracts and the theory of the firm. The Quarterly Journal of Economics 117, 39–84. doi:10.1162/003355302753399445.

Balakrishnan, R., Labro, E., Soderstrom, N.S., 2014. Cost structure and sticky costs. Journal of Management Accounting Research 26, 91–116. doi:10.2308/jmar-50831.

Bandiera, O., Prat, A., Valletti, T., 2009. Active and passive waste in government spending: Evidence from a policy experiment. American Economic Review 99, 1278–1308. doi:10.1257/aer.99.4.1278.

Banker, R., Huang, R., Li, X., Yan, Y., 2024a. Strategy typology and cost structure: A textual analysis approach. Production and Operations Management 33, 1285–1305. doi:10.1177/10591478241245144.

Banker, R.D., Byzalov, D., Chen, L.T., 2013a. Employment protection legislation, adjustment costs and cross-country differences in cost behavior. Journal of Accounting and Economics 55, 111–127. doi:10.1016/j.jacceco.2012.08.003.

Banker, R.D., Byzalov, D., Plehn-Dujowich, J.M., 2014. Demand uncertainty and cost behavior. The Accounting Review 89, 839–865. doi:10.2308/accr-50661.

Banker, R.D., Byzalov, D., Threinen, L., 2013b. Determinants of international differences in asymmetric cost behavior. SSRN Electronic Journal. doi:10.2139/ssrn.2312772.

Banker, R.D., Chen, L.T., 2006. Predicting earnings using a model based on cost variability and cost stickiness. The Accounting Review 81, 285–307. doi:10.2308/accr.2006.81.2.285.

Banker, R.D., Huang, R., Wang, Y., Yan, Y., 2024b. Flexible or rigid? Evidence on managerial ability and cost structure. Contemporary Accounting Research doi:10.1111/1911-3846.13003.

Bartik, T.J., 1991. Who Benefits from State and Local Economic Development Policies? W.E. Upjohn Institute. doi:10.17848/9780585223940.

Belina, H., Surysekar, K., Weismann, M., 2019. On the medical loss ratio (MLR) and sticky selling general and administrative costs: Evidence from health insurers. Journal of Accounting and Public Policy 38, 53–61. doi:10.1016/j.jaccpubpol.2019.01.004.

Celli, V., 2022. Causal mediation analysis in economics: Objectives, assumptions, models. Journal of Economic Surveys 36, 214–234. doi:10.1111/joes.12452.

Chemin, M., 2020. Judicial efficiency and firm productivity: Evidence from a world database of judicial reforms. The Review of Economics and Statistics 102, 49–64. doi:10.1162/rest_a_00799.

Chen, S., Chao, X., Wang, C., Xue, Z., 2025. Government digital product procurement and enterprise digital transformation. China & World Economy 33, 139–166. doi:10.1111/cwe.12605.

Cheng, H., Fan, H., Hoshi, T., Hu, D., 2019. Do Innovation Subsidies Make Chinese Firms More Innovative? Evidence from the China Employer Employee Survey. NBER Working Paper 25432. National Bureau of Economic Research. Cambridge, MA. doi:10.3386/w25432.

Choi, J., Penciakova, V., Saffie, F., 2024. Political Connections, Allocation of Stimulus Spending, and the Jobs Multiplier. NBER Working Paper 32574. National Bureau of Economic Research. Cambridge, MA. doi:10.3386/w32574.

Clemens, J., Rogers, P., 2026. Demand shocks, procurement policies, and the nature of medical innovation: Evidence from wartime prosthetic device patents. The Review of Economics and Statistics 108, 75–89. doi:10.1162/rest_a_01365.

Cooper, M., Knott, A.M., Yang, W., 2022. RQ innovative efficiency and firm value. Journal of Financial and Quantitative Analysis 57, 1649–1694. doi:10.1017/S0022109021000417.

Cooper, R.W., Haltiwanger, J.C., 2006. On the nature of capital adjustment costs. Review of Economic Studies 73, 611–633. doi:10.1111/j.1467-937X.2006.00389.x.

Cui, Y., Che, W., Liu, T., Qin, B., Yang, Z., 2021. Pre-training with whole word masking for Chinese BERT. IEEE/ACM Transactions on Audio, Speech, and Language Processing 29, 3504–3514. doi:10.1109/TASLP.2021.3124365.

Dai, X., Li, Y., Chen, K., 2021. Direct demand-pull and indirect certification effects of public procurement for innovation. Technovation 101, 102198. doi:10.1016/j.technovation.2020.102198.

Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), Association for Computational Linguistics, Minneapolis, Minnesota. pp. 4171–4186. doi:10.18653/v1/N19-1423.

DeWenter, K.L., Malatesta, P.H., 2001. State-owned and privately owned firms: An empirical analysis of profitability, leverage, and labor intensity. American Economic Review 91, 320–334. doi:10.1257/aer.91.1.320.

Dierynck, B., Landsman, W.R., Renders, A.H.K., 2012. Do managerial incentives drive cost behavior? Evidence about the role of the zero earnings benchmark for labor cost behavior in private Belgian firms. The Accounting Review 87, 1219–1246. doi:10.2308/accr-50153.

Edler, J., Georghiou, L., 2007. Public procurement and innovation—Resurrecting the demand side. Research Policy 36, 949–963. doi:10.1016/j.respol.2007.03.003.

Faccio, M., 2006. Politically connected firms. American Economic Review 96, 369–386. doi:10.1257/000282806776157704.

Farrell, J., Klemperer, P., 2007. Coordination and lock-in: Competition with switching costs and network effects, in: Armstrong, M., Porter, R.H. (Eds.), Handbook of Industrial Organization. Elsevier. volume 3, pp. 1967–2072. doi:10.1016/S1573-448X(06)03031-7.

Farrell, J., Saloner, G., 1985. Standardization, compatibility, and innovation. Rand Journal of Economics 16, 70–83. doi:10.2307/2555589.

Fisman, R., Svensson, J., 2007. Are corruption and taxation really harmful to growth? Firm level evidence. Journal of Development Economics 83, 63–75. doi:10.1016/j.jdeveco.2005.09.009.

Gentzkow, M., Kelly, B., Taddy, M., 2019. Text as data. Journal of Economic Literature 57, 535–574. doi:10.1257/jel.20181020.

Georghiou, L., Edler, J., Uyarra, E., Yeow, J., 2014. Policy instruments for public procurement of innovation: Choice, design and assessment. Technological Forecasting and Social Change 86, 1–12. doi:10.1016/j.techfore.2013.09.018.

Goldsmith-Pinkham, P., Sorkin, I., Swift, H., 2020. Bartik instruments: What, when, why, and how. American Economic Review 110, 2586–2624. doi:10.1257/aer.20181047.

Grandia, J., Volker, L., 2023. Public Procurement Theory, Practices and Tools. Palgrave Macmillan, Cham. doi:10.1007/978-3-031-18490-1.

Grimmer, J., Stewart, B.M., 2013. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. Political Analysis 21, 267–297. doi:10.1093/pan/mps028.

Gross, D.P., Sampat, B.N., 2023. America, jump-started: World war II R&D and the takeoff of the US innovation system. American Economic Review 113, 3323–3356. doi:10.1257/aer.20221365.

Guerzoni, M., Raiteri, E., 2015. Demand-side vs. supply-side technology policies: Hidden treatment and new empirical evidence on the policy mix. Research Policy 44, 726–747. doi:10.1016/j.respol.2014.10.009.

Gulen, H., Ion, M., 2016. Policy uncertainty and corporate investment. The Review of Financial Studies 29, 523–564. doi:10.1093/rfs/hhv050.

Hanlon, M., Lester, R., Verdi, R., 2015. The effect of repatriation tax costs on U.S. multinational investment. Journal of Financial Economics 116, 179–196. doi:10.1016/j.jfineco.2014.12.004.

Hartlieb, S., Loy, T.R., 2022. The impact of cost stickiness on financial reporting: evidence from income smoothing. Accounting & Finance 62, 3913–3950. doi:10.1111/acfi.12910.

He, J., Tian, X., Yang, H., Zuo, L., 2020. Asymmetric cost behavior and dividend policy. Journal of Accounting Research 58, 989–1021. doi:10.1111/1475-679X.12328.

Holzhacker, M., Krishnan, R., Mahlendorf, M.D., 2015. The impact of changes in regulation on cost behavior. Contemporary Accounting Research 32, 534–566. doi:10.1111/1911-3846.12082.

Howell, S., Rathje, J., Van Reenen, J., Wong, J., 2021. Opening up Military Innovation: Causal Effects of Reforms to U.S. Defense Research. NBER Working Paper 28700. National Bureau of Economic Research. Cambridge, MA. doi:10.3386/w28700.

Hu, A.G., Zhang, P., Zhao, L., 2017. China as number one? Evidence from china's most recent patenting surge. Journal of Development Economics 124, 107–119. doi:10.1016/j.jdeveco.2016.09.004.

Hui, L., Xie, H., Chen, X., 2024. Digital technology, the industrial internet, and cost stickiness. China Journal of Accounting Research 17, 100339. doi:10.1016/j.cjar.2023.100339.

Imai, K., Keele, L., Tingley, D., 2010. A general approach to causal mediation analysis. Psychological Methods 15, 309–334. doi:10.1037/a0020761.

Kallapur, S., Eldenburg, L., 2005. Uncertainty, real options, and cost behavior: Evidence from Washington State hospitals. Journal of Accounting Research 43, 735–752. doi:10.1111/j.1475-679X.2005.00188.x.

Kantor, S., Whalley, A., 2025. Moonshot: Public R&D and growth. American Economic Review 115, 2891–2925. doi:10.1257/aer.20220540.

Klemperer, P., 1987. Markets with consumer switching costs. The Quarterly Journal of Economics 102, 375–394. doi:10.2307/1885068.

Krieger, B., Prüfer, M., Strecke, L., 2024. Public procurement can hinder innovation. SSRN Electronic Journal. doi:10.2139/ssrn.4774494.

Krieger, B., Zipperer, V., 2022. Does green public procurement trigger environmental innovations? Research Policy 51, 104516. doi:10.1016/j.respol.2022.104516.

Lewbel, A., 2012. Using heteroscedasticity to identify and estimate mismeasured and endogenous regressor models. Journal of Business & Economic Statistics 30, 67–80. doi:10.1080/07350015.2012.643126.

Li, Y., Feng, P., Qi, T., Yan, J., Huang, Y., 2024. Enterprise digital transformation, managerial myopia and cost stickiness. Humanities and Social Sciences Communications 11, 1389. doi:10.1057/s41599-024-03926-1.

Li, Y., Georghiou, L., 2016. Signaling and accrediting new technology: Use of procurement for innovation in China. Science and Public Policy 43, 338–351. doi:10.1093/scipol/scv044.

Lichtenberg, F.R., 1988. The private R&D investment response to federal design and technical competitions. The American Economic Review 78, 550–559.

Macchiavello, R., Morjaria, A., 2023. Relational contracts: Recent empirical advancements and open questions. Journal of Institutional and Theoretical Economics 179, 673–700. doi:10.1628/jite-2023-0048.

Manso, G., 2011. Motivating innovation. The Journal of Finance 66, 1823–1860. doi:10.1111/j.1540-6261.2011.01688.x.

McGuire, S.T., Rane, S.G., Weaver, C.D., 2023. Cost structure and tax-motivated income shifting. The Accounting Review 98, 435–456. doi:10.2308/TAR-2020-0190.

Mitton, T., 2024. Economic significance in corporate finance. The Review of Corporate Finance Studies 13, 38–79. doi:10.1093/rcfs/cfac008.

Nickell, S.J., 1996. Competition and corporate performance. Journal of Political Economy 104, 724–746. doi:10.1086/262040.

Novy-Marx, R., 2011. Operating leverage. Review of Finance 15, 103–134. doi:10.1093/rof/rfq019.

Obwegeser, N., Müller, S.D., 2018. Innovation and public procurement: Terminology, concepts, and applications. Technovation 74–75, 1–17. doi:10.1016/j.technovation.2018.02.015.

OECD, 2017. Public Procurement for Innovation: Good Practices and Strategies. OECD Public Governance Reviews, OECD. doi:10.1787/9789264265820-en.

Seitz, M., Watzinger, M., 2017. Contract enforcement and R&D investment. Research Policy 46, 182–195. doi:10.1016/j.respol.2016.09.015.

Shleifer, A., Vishny, R.W., 1994. Politicians and firms. The Quarterly Journal of Economics 109, 995–1025. doi:10.2307/2118354.

Slavtchev, V., Wiederhold, S., 2016. Does the technological content of government demand matter for private R&D? Evidence from US states. American Economic Journal: Macroeconomics 8, 45–84. doi:10.1257/mac.20130069.

Tan, W., Liu, Y., Shen, M., 2024. Supporting hand: Government digital procurement and enterprise digital technology innovation. Journal of Shanghai University of Finance and Economics 26, 18–32. doi:10.16538/j.cnki.jsufe.2024.03.002. (In Chinese).

Tang, W., Wang, Y., Wu, M., 2025. Local favoritism in China's public procurement: Information frictions or incentive distortion? Journal of Urban Economics 145, 103716. doi:10.1016/j.jue.2024.103716.

Teece, D.J., 1986. Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. Research Policy 15, 285–305. doi:10.1016/0048-7333(86)90027-2.

United Nations Conference on Trade and Development, 2019. Digital Economy Report 2019: Value Creation and Capture : Implications for Developing Countries. Technical Report. United Nations. Geneva.

Uyarra, E., Zabala-Iturriagagoitia, J.M., Flanagan, K., Magro, E., 2020. Public procurement, innovation and industrial policy: Rationales, roles, capabilities and implementation. Research Policy 49, 103844. doi:10.1016/j.respol.2019.103844.

Wang, R., Wang, X., Xu, G., Zha, T., 2024. Privatization's Impacts on State-Owned Enterprises: A Tale of Zombie versus Healthy Firms. NBER

Working Paper 32795. National Bureau of Economic Research. Cambridge, MA. doi:10.3386/w32795.

World Bank, 2023. Digital Progress and Trends Report 2023. Technical Report. World Bank. Washington, DC.

World Bank, 2025. Digital Progress and Trends Report 2025: Strengthening AI Foundations. Technical Report. World Bank. Washington, DC.

Zhang, Y., Yang, Y., 2025. Can governmental innovation subsidies promote firm innovative economic efficiency? Evidence from China. International Review of Financial Analysis 106, 104473. doi:10.1016/j.irfa.2025.104473.